

CS 320: Software Metrics Assignment (Function Point Analysis) - Gayatri Marathe

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Project Name: UMass Marketplace

Objective: Function Points for the testing related features in the backend (Authentication, Listings, and Search/Filter):

For our current UMass Marketplace backend, the main Internal Logical Files (ILFs) are:

- User
- Listing
- Product
- Category

We used 0 External Interface Files (EIFs) for these functions because the current uploaded backend only uses its own internal models and does not show external user identifiable files maintained by another application.

1) Register

- **ILF:** User
- **EIF:** None
- **EI:** user sends registration data
- **EO:** system returns token + user
- **EQ:** none

Information Domain Value	Count	Weight (Average)	Total
External Inputs (EIs)	1	4	4
External Outputs (EOs)	1	5	5
External Inquiries (EQs)	0	4	0
Internal Logical Files (ILFs)	1	10	10
External Interface Files (EIFs)	0	7	0
Count total			19

2) Login

- **ILF:** User
- **EIF:** None
- **EI:** user submits login credentials
- **EO:** system returns token + user
- **EQ:** none

Information Domain Value	Count	Weight (Average)	Total
External Inputs (EIs)	1	4	4
External Outputs (EOs)	1	5	5
External Inquiries (EQs)	0	4	0
Internal Logical Files (ILFs)	1	10	10
External Interface Files (EIFs)	0	7	0
Count total			19

3) Create Listing

- **ILFs:** Listing, User, Product, Category
- **EIF:** None
- **EI:** seller submits listing data
- **EO:** system returns created listing
- **EQ:** none

Information Domain Value	Count	Weight (Average)	Total
External Inputs (EIs)	1	4	4
External Outputs (EOs)	1	5	5
External Inquiries (EQs)	0	4	0
Internal Logical Files (ILFs)	4	10	40

External Interface Files (EIFs)	0	7	0
Count total			49

4) Browse/Search Listings

- **ILFs**: Listing, User, Product, Category
- **EIF**: None
- **EI**: none
- **EO**: system returns filtered/paginated listings
- **EQ**: none

Information Domain Value	Count	Weight (Average)	Total
External Inputs (EIs)	1	4	0
External Outputs (EOs)	1	5	5
External Inquiries (EQs)	0	4	0
Internal Logical Files (ILFs)	4	10	40
External Interface Files (EIFs)	0	7	0
Count total			45

Total Function Points:

Function	Function Point Total
Register	19
Login	19
Create Listing	49
Browse/Search Listings	45
Total	132